

SQL & MySQL Roadmap: From Basics to Advanced

SQL & MySQL Basics

- What is SQL and MySQL?
- Installing MySQL and MySQL Workbench
- Databases vs Tables
- Basic SQL Syntax
- CREATE, DROP, and ALTER DATABASE
- CREATE, DROP, and ALTER TABLE
- Data Types in MySQL
- INSERT INTO Statement
- SELECT Statement
- WHERE Clause
- UPDATE Statement
- DELETE Statement
- DISTINCT Keyword
- ORDER BY Clause
- LIMIT Clause
- Aliases (AS)
- Basic Constraints (NOT NULL, UNIQUE, DEFAULT)

Joins & Relationships

- Primary Key and Foreign Key
- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL OUTER JOIN
- SELF JOIN
- CROSS JOIN
- Joining Multiple Tables
- ON vs USING in JOINS

Intermediate SQL

- GROUP BY and HAVING
- Aggregate Functions (SUM, COUNT, AVG, MIN, MAX)
- IN, BETWEEN, LIKE
- IS NULL & IS NOT NULL
- UNION and UNION ALL
- Subqueries (Single and Multiple Row)
- EXISTS and NOT EXISTS
- CASE Statement
- String Functions (CONCAT, LENGTH, SUBSTRING, etc.)
- Numeric Functions (ROUND, CEIL, FLOOR)
- Date and Time Functions
- IFNULL & COALESCE

Advanced SQL

- Views
- Stored Procedures
- Stored Functions
- Triggers
- Indexes (Single and Composite)
- Transactions (START, COMMIT, ROLLBACK)
- Locking (Shared and Exclusive Locks)
- Window Functions (ROW_NUMBER, RANK, DENSE_RANK)
- CTE (Common Table Expressions)
- Recursive Queries
- Performance Tuning with EXPLAIN
- Temporary Tables

Database Design & Modeling

- Normalization (1NF, 2NF, 3NF)
- Denormalization
- ER Diagrams
- One-to-One Relationships
- One-to-Many Relationships
- Many-to-Many Relationships

- Schema Design Best Practices

MySQL Tools & Administration

- MySQL Workbench Overview
- Importing & Exporting Data
- User Management & Privileges
- Backup and Restore
- MySQL Configuration Files
- Connection Pooling
- Working with CLI
- MySQL Security Basics
- Replication Concepts
- Monitoring MySQL Performance